A Correlation-Aware Graph Transformer for Joint Regional Load and Operational Stress Forecasting in Bangladesh

MD. Nafiz Ahmed Tanim
Dept. of CSE, AIUB
Dhaka, Bangladesh
24-56644-1@student.aiub.edu

MD. Tanvir Hasan
Dept. of CSE, AIUB
Dhaka, Bangladesh
23-54796-3@student.aiub.edu

MD. Robiul Islam
Dept. of CSE, AIUB
Dhaka, Bangladesh
24-56643-1@student.aiub.edu

Wasimul Bari Tonmoy
Dept. of CSE, AIUB
Dhaka, Bangladesh
24-56653-1@student.aiub.edu

MD. Hasanuzzaman
Dept. of CSE, AIUB
Dhaka, Bangladesh
24-56666-1@student.aiub.edu

Dipta Justin Gomes
Dept. of CSE, AIUB
Dhaka, Bangladesh
diptagomes@aiub.edu

***Abstract*—Day-ahead planning in developing-economy national grids requires synchronised visibility over regional electricity demand and system-wide operational pressure, yet load prediction and stress monitoring are often handled separately. On the Bangladesh national power system, daily records from nine administrative divisions jointly report evening-peak load with limitation, reserve, and shedding indicators, making coupled forecasting necessary. We implement the Correlation-Aware Multi-Task Forecasting Framework as a Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT) that encodes a seven-day lookback with a train-estimated correlation graph and decodes regional demand and a bounded Operational Stress Index (OSI) through shared spatio-temporal encoding and task-specific heads. On a frozen held-out test year, correlation-graph PF-STGT (S2) achieves the strongest joint performance in the internal suite—macro demand MAE 88.65 MW and OSI R^2 0.745—with statistically supported demand improvement over the historical hybrid reference (median ΔMAE −5.43 MW, $p = 5.5 \times 10^{-5}$). Ablation favours correlation adjacency over geographical or hybrid priors and documents a demand-only versus joint-stress trade-off. Post-hoc attribution highlights limitation and calendar coalitions with partial cross-method agreement. Results are scoped to offline chronological evaluation on Bangladesh division-level data rather than demonstrated deployment outcomes.**

***Index Terms*—Multi-task learning, spatio-temporal forecasting, operational stress index, correlation graph, smart grid, Bangladesh**

I. Introduction

National grid operators must anticipate regional load and assess system pressure on a day-ahead horizon to coordinate generation dispatch, reserve planning, and constraint management. In developing economies, supply limitations and load shedding often coincide with rapid demand growth, so planners need forecasts that show both where load will concentrate and how severely operational constraints may bind system-wide. On the Bangladesh national power system, operational data are reported across nine administrative divisions with regional evening-peak demand and limitation dynamics co-visible in each daily record [1].

Despite these data, forecasting practice often treats load prediction and stress monitoring separately. Classical and ensemble models [2]–[4] forecast demand without explicit inter-division graph structure; shallow recurrent or graph-convolution architectures [5], [6] may fail to combine long-range temporal context with informative multi-node representations. Graph-based methods from residential or microgrid settings often rely on geographical adjacency that may not capture data-driven co-movement among regional loads [7]. Operational stress—from reserve shortfall, limitation pressure, and shedding intensity—is seldom forecast jointly with regional demand; shedding studies [8] typically address control rather than day-ahead graded stress indices. Explainability is rarely integrated with multi-node graph-temporal models that emit coupled load and stress outputs [9], [10].

We formulate day-ahead forecasting as a supervised multi-task problem [11] over nine spatial units. Task 1 predicts next-day regional evening-peak demand (MW); Task 2 predicts a graph-level Operational Stress Index (OSI) in $[0, 1]$ summarising system-wide pressure from shedding intensity, generation reserve margin, and operational limitation components. Demand and OSI are operationally related yet statistically non-redundant: demand describes expected load by region, whereas OSI encodes constraint-related pressure not fully determined by demand alone.

This paper proposes the Correlation-Aware Multi-Task Forecasting Framework, implemented as Parallel-Fusion Spatio-Temporal Graph Transformer (PF-STGT). The framework ingests a seven-day lookback of node-level and national context features with a train-estimated correlation graph, combines graph-aware spatial attention with temporal transformer capacity [12], and decodes regional demand and bounded OSI through task-specific heads with a repaired multi-task loss.

Evaluation on chronologically partitioned Bangladesh grid records includes classical, recurrent, and graph-convolution benchmarks, configuration ablations, inferential testing, and post-hoc attribution.

The principal contributions are: (i) a correlation-graph PF-STGT for joint day-ahead regional demand and OSI prediction on Bangladesh national grid data; (ii) evidence-driven architecture selection favouring correlation-only adjacency over geographical or hybrid priors; (iii) a repaired multi-task training protocol with explicit demand–stress loss balancing; (iv) an empirical evaluation programme with benchmarks, ablations, Wilcoxon testing with Bonferroni correction, and bootstrap confidence intervals; and (v) integrated explainability with multi-method attribution for both tasks. Section II presents the methodology; Sections III–IV report experiments and results; Sections V–VI discuss findings and conclude.

II. Methodology

A. Problem Formulation

Let $\mathcal{V} = \{1, \ldots, N\}$ denote $N{=}9$ administrative divisions. At forecast origin t, the model receives a lookback window of $T{=}7$ days and predicts targets at horizon $H{=}1$. Node features $\mathbf{X}^{\text{node}} \in \mathbb{R}^{T \times N \times F_n}$ ($F_n{=}9$) encode division-level demand, supply, load, lags, rolling means, and regional shares. Global features $\mathbf{X}^{\text{global}} \in \mathbb{R}^{T \times F_g}$ ($F_g{=}17$) encode calendar indicators, grid aggregates, limitation variables, and national generation scalars. Task 1 outputs regional demand $\hat{\mathbf{y}}^{\text{demand}} \in \mathbb{R}^N$ (MW); Task 2 outputs graph-level OSI $\hat{y}^{\text{osi}} \in [0, 1]$.

OSI integrates three train-normalised components:

$$c_1 = L_{\text{total}}/D_{\text{total}}, \quad c_2 = 1 - GR/\text{Highest\_Gen}, \quad c_3 = \text{TOL}/\text{Highest\_Gen},$$
$$\text{OSI} = \text{mean}\big(\text{minmax}_{\text{train}}(c_1),\, \text{minmax}_{\text{train}}(c_2),\, \text{minmax}_{\text{train}}(c_3)\big). \tag{1}$$

where L_{total}, D_{total}, GR, and TOL denote total shedding, demand, generation reserve, and operational limitation. Same-day OSI is excluded from inputs when forecasting OSI($t{+}1$) to prevent leakage.

The joint objective combines macro-averaged Huber loss on demand [13] with MSE on OSI:

$$\mathcal{L} = \mathcal{L}_{\text{demand}}^{\text{raw}}/100 + \lambda_2\, \mathcal{L}_{\text{osi}}, \quad \lambda_2 = 20. \tag{2}$$

Demand-loss scaling prevents collapse to demand-only learning when OSI targets lie on $[0, 1]$.

B. Dataset and Graph Construction

We use the Bangladesh Smart Grid dataset (Mendeley Data, Version 5) [1]: 1,850 daily records (Nov. 2019–Dec. 2024) across nine divisions. Chronological partitions yield 1,281 training, 263 validation, and 264 test sliding windows (Table I). All preprocessing parameters—scaling, encoding, graph weights, OSI bounds—are estimated on training data only.

Pairwise Pearson correlations ρ_{ij} on training-partition evening-peak demand define undirected edges with $\rho_{ij} \geq 0.65$; weights equal ρ_{ij} and are row-normalised to form adjacency $\mathbf{A}$. This correlation-only graph (33 of 36 pairs retained) replaces geographical and hybrid priors following ablation evidence. Geographical adjacency (21 pairs) and hybrid adjacency (24 pairs) serve as ablation comparators only.

TABLE I
Dataset and S2 training configuration.

Dataset	Value	Training	Value
Divisions (N)	9	Optimiser	Adam, lr $5{\times}10^{-4}$
Window (T) / horizon	7 / 1 day	Batch / epochs	32 / $\leq$200
Node / global features	9 / 17	λ_2 (Eq. 2)	20.0
Train / val / test windows	1281 / 263 / 264	Parameters	749,058
Graph edges ($\tau{=}0.65$)	33 / 36 pairs	Early stop	Patience 15

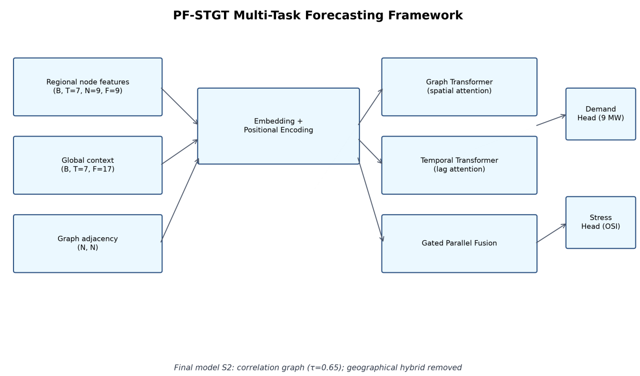

Fig. 1. End-to-end PF-STGT multi-task forecasting framework.

C. PF-STGT Architecture

Fig. 1 shows the end-to-end pipeline; Fig. 2 details configuration S2. Node and global tensors are embedded with regional identity and positional encoding. A *spatial branch* applies graph-transformer self-attention over nodes at each timestep with additive bias from $\mathbf{A}$:

$$\text{Attn}_{\text{spatial}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B}_{\text{adj}}\right)\mathbf{V}. \tag{3}$$

A *temporal branch* applies a shared transformer encoder over each node's seven-day sequence. Learnable gating fuses the branches; the final timestep yields shared node embeddings $\mathbf{H}_{\text{shared}}$. A per-node linear head decodes regional demand; a graph-level head combining pooled node states and global context decodes OSI through a bounded activation. Both tasks share encoders and differ only in decoding layers.

III. Experimental Setup

A. Baselines and Metrics

We compare S2 against seven reference models spanning classical machine learning (linear regression, random forest [2], XGBoost [3]), recurrent deep learning (LSTM [5], GRU [14]), graph convolution (T-GCN [6]), and the historical PF-STGT reference B07 (hybrid geographical–correlation graph, W20 protocol). All models share chronological partitions, the frozen feature store, windowing ($T{=}7$, $H{=}1$), and leakage policies.

Task 1 is evaluated with macro-averaged MAE (primary metric), RMSE, MAPE, and R^2 over nine divisions. Task 2

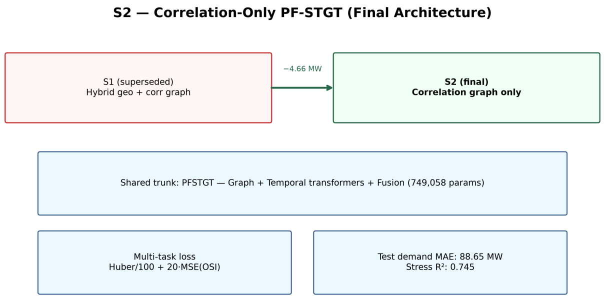

Fig. 2. Configuration S2: correlation-only graph PF-STGT.

uses MAE and R^2 on OSI. Paired Wilcoxon signed-rank tests [15] on per-window macro demand MAE apply Bonferroni correction [16]; bootstrap 95% confidence intervals [17] and Cohen's d [18] supplement p-values.

B. Training Configuration

S2 trains with Adam [19] ($\eta{=}5{\times}10^{-4}$, weight decay 10^{-4}), batch size 32, up to 200 epochs with early stopping (patience 15 on composite validation score), and gradient clipping at 1.0. Model capacity: $d_{\text{model}}{=}128$, four heads, two spatial and two temporal layers (749,058 parameters). Deep baselines use AdamW with up to 60 epochs; classical models use training-partition standardisation only. Seed 42 governs all stochastic operations.

C. Ablation Design

Six PF-STGT configurations (A1–A6) isolate graph topology (none, geographical, hybrid, correlation-only), transformer removal, and single-task demand-only training. A1 equals B07; A6 equals S2. Architecture simplification variants S1–S4 further test component removal on the correlation graph.

IV. Results

A. Benchmark Performance

Table II and Fig. 3 summarise held-out test performance ($n{=}264$ windows). S2 attains the lowest macro demand MAE (88.65 MW; RMSE 127.29 MW; MAPE 6.55%) and highest OSI R^2 (0.745; MAE 0.0371), improving 4.66 MW (5.0%) over B07 (93.31 MW) and 8.38 MW over random forest B02 (97.03 MW). Recurrent baselines B04/B05 exceed 233 MW MAE with negative macro R^2; T-GCN B06 reaches 257.21 MW. Classical ensembles remain competitive but do not match S2 megawatt accuracy despite B02's higher macro R^2 (0.984).

Paired Wilcoxon tests confirm benchmark rankings. B07 significantly outperforms B01–B06 at Bonferroni $\alpha{=}0.0083$. S2 versus B07 yields median ΔMAE $= -5.43$ MW, $p{=}5.5{\times}10^{-5}$, bootstrap 95% CI $[-7.17, -2.16]$ MW—the only graph-variant comparison with statistically supported demand improvement over the historical reference.

TABLE II
BENCHMARK COMPARISON AND WILCOXON TESTS (TEST SET, $n{=}264$).

ID	Model	MAE	RMSE	MAPE	R^2	OSI MAE	OSI R^2
S2	Corr.-Only PF-STGT	88.65	127.29	6.55	0.684	0.0371	0.745
B07	PF-STGT W20 hybrid	93.31	128.81	6.76	0.674	0.0499	0.585
B02	Random Forest	97.03	156.99	7.04	0.984	0.0481	0.555
B03	XGBoost	109.73	178.53	7.99	0.979	0.0497	0.525
B06	T-GCN	257.21	301.06	15.72	−0.483	0.0891	−0.304

Wilcoxon (demand MAE): S2 vs B07: median $\Delta{=}-5.43$ MW, $p{=}5.5{\times}10^{-5}$, CI $[-7.17, -2.16]$

TABLE III
ABLATION STUDY (TEST SET). A6 = S2.

ID	Variant	MAE	R^2	OSI MAE	OSI R^2
A4	Single-Task	86.89	0.731	–	–
A6 (S2)	Corr. Graph	88.65	0.684	0.0371	0.745
A1	PF-STGT W20	93.31	0.674	0.0499	0.585
A5	Geo Graph	97.98	0.554	0.0340	0.764

B. Regional and Stress Analysis

Per-region MAE under S2 spans 32.73 MW (Barishal) to 293.98 MW (Dhaka). Dhaka accounts for ∼35.7% of national demand and dominates the megawatt error budget; S2 reduces Dhaka MAE by 5.80 MW versus B07 while peripheral divisions remain below 60 MW. Macro summaries understate planning significance for the principal load centre.

On OSI, S2 reduces MAE by 0.0128 and raises R^2 by 0.160 absolute versus B07. Deep baselines B04–B06 yield negative stress R^2 (-0.191 to -0.304). No paired Wilcoxon tests were computed for OSI; stress claims rest on point estimates.

C. Ablation Study

Table III and Fig. 4 report configuration results. Correlation-graph A6 (S2) delivers the lowest demand MAE among multi-task variants (88.65 MW) with OSI R^2 0.745. Geographical-graph A5 attains marginally better stress metrics (MAE 0.0340, R^2 0.764) but 9.33 MW higher demand error. Demand-only A4 achieves the lowest demand MAE (86.89 MW) without OSI output—a 1.76 MW advantage over S2 documenting a genuine demand–stress Pareto trade-off. Transformer removal on the hybrid graph (A3) is not statistically distinguishable from A1 ($p{=}0.384$); geographical graph A5 significantly degrades demand ($p{=}1.48{\times}10^{-4}$).

D. Explainability

Post-hoc attribution [10], [20] on the frozen S2 checkpoint applies grouped GradientSHAP, permutation importance, and spatial attention export. On stress, limitation stack (G8) and calendar/trend (G6) coalitions rank highest ($|\phi|{=}0.0191$ and 0.0190). On Dhaka demand, calendar (G6), engineered lags (G4), and national generation (G10) dominate ($|\phi|{=}162.34$, 101.26, 91.44). Fig. 5 shows node-level attribution mass concentrated on Dhaka (340.36), Rajshahi, and Khulna; Spearman correlation between spatial attention and correlation adjacency is $\rho{=}0.422$. Dual-path comparison of gradient attributions with OSI component drivers agrees in 13 of 24 case studies (52.2%); demand gradient and permutation rankings disagree ($\rho{=}-0.564$).

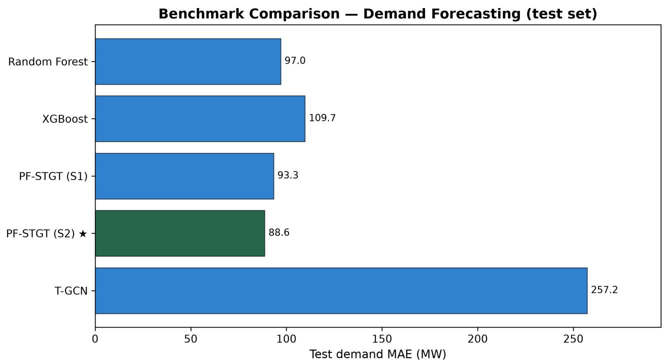

Fig. 3. Test-set macro demand MAE benchmark comparison.

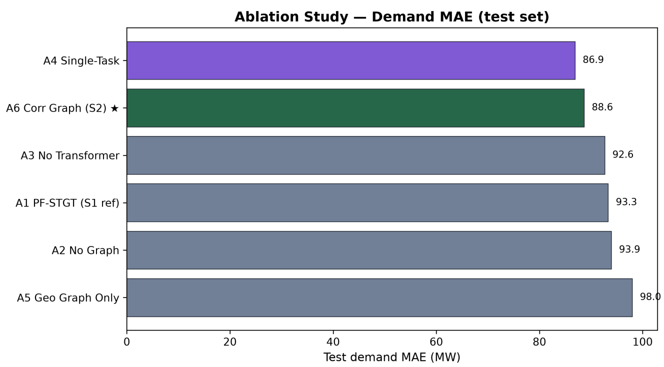

Fig. 4. Ablation study demand MAE on the test set.

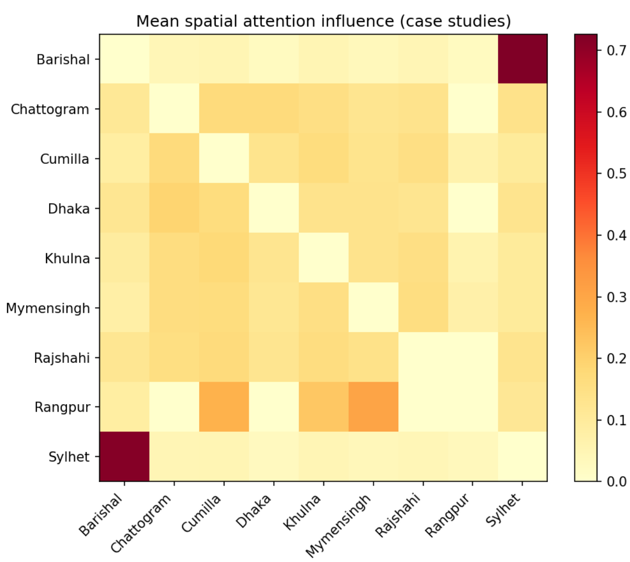

Fig. 5. Node-level attribution heatmap (S2 checkpoint).

V. Discussion

The frozen evaluation demonstrates that one correlation-graph PF-STGT can supply operationally coupled day-ahead load and stress visibility from shared representations on Bangladesh national grid data. Three design lessons emerge from the evidence chain.

Graph prior selection. Train-estimated correlation adjacency outperforms geographical-only and hybrid priors on joint demand accuracy, while geographical conditioning significantly degrades demand (p=1.48×10^{-4}). Inter-regional coupling on this national footprint reflects shared growth and seasonality more than border adjacency—a pattern under-represented in US/European graph-forecasting benchmarks [21], [22].

Multi-task trade-offs. Repaired loss balancing (λ_2=20, demand scaling by 100 MW) restores stress learnability absent in earlier configurations. Demand-only training attains a lower megawatt ceiling, but multi-task S2 enables informative OSI outputs (R^2=0.745) at a modest 1.76 MW demand offset—the empirically preferred joint optimum for synchronised planning.

Regional error geography. Error concentrates in Dhaka across all model families, indicating structural difficulty at the dominant load centre rather than graph-conditioning artefact. Operators should report the capital division separately while relying on S2 for peripheral and mid-scale service.

Limitations. Evidence is bounded to a single Bangladesh timeline with daily cadence and frozen internal benchmarks; no external replication or deployment trial is included. Inferential support applies to demand MAE; OSI gains lack hypothesis tests. Attribution describes model behaviour, not physical causation, with documented cross-method discordance. Uncertainty quantification and S2 residual diagnostics were not executed. Practical implications are planning diagnostics grounded in offline accuracy—not demonstrated improvements in grid reliability or economic outcome.

VI. Conclusion

This study investigated whether a single spatio-temporal framework can jointly forecast next-day regional evening-peak demand and a graph-level Operational Stress Index from Bangladesh national grid records under chronological evaluation. The Correlation-Aware Multi-Task Forecasting Framework—implemented as correlation-graph PF-STGT with parallel spatial and temporal transformer encoding and repaired multi-task loss—delivers the strongest joint credentials in the frozen benchmark suite: macro demand MAE 88.65 MW and OSI R^2 0.745, with statistically supported demand improvement over the historical hybrid reference.

Five outcomes define the contribution: coupled day-ahead load and stress forecasts from one checkpoint; correlation adjacency outperforming geographical priors; a documented demand–stress trade-off against demand-only training; predictable regional error concentration with Dhaka gains; and multi-method attribution with explicit discordance reporting. The work extends spatio-temporal forecasting to coupled stress indices on a previously unreported Bangladesh division-level case [1], demonstrating that graph priors and task weighting require empirical tuning rather than domain intuition. Coupled day-ahead load-and-stress forecasting is achievable as a single-model service when graph structure, multi-task calibration, and division-level monitoring are co-designed with the evaluation evidence—appropriately qualified as offline evaluation rather than operational deployment.

ACKNOWLEDGMENT

The authors thank Dipta Justin Gomes (AIUB) for guidance and supervision throughout this research.

REFERENCES

[1] M. I. Hossain, T. Nafs, and S. Yeaser, "Structured dataset of daily electricity demand, generation, load shedding, and supply constraints in bangladesh (2019–2024)," 2025, version 5; accessed for model training and evaluation. [Online]. Available: https://data.mendeley.com/datasets/x7r7wdb39k

[2] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[3] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.

[4] H. Musbah and M. Elsaraiti, "Forecasting load consumption: a comprehensive evaluation of deep learning and machine learning techniques," *Electric Power Systems Research*, vol. 247, p. 111834, 2025.

[5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[6] L. Zhao, Y. Song, C. Zhang, Y. Liu, P. Wang, T. Lin, H. Deng, and H. Li, "T-GCN: A temporal graph convolutional network for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 9, pp. 3848–3858, 2020.

[7] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. 5th Int. Conf. Learning Representations (ICLR)*, 2017.

[8] G. Paphitis and M. Panteli, "A two-stage machine learning framework for predicting spatial load shedding in cascading power system failures," in *2025 IEEE Kiel PowerTech*. IEEE, 2025, pp. 1–6.

[9] L. Baur, K. Ditschuneit, M. Schambach, C. Kaymakci, T. Wollmann, and A. Sauer, "Explainability and interpretability in electric load forecasting using machine learning techniques – a review," *Energy and AI*, vol. 16, p. 100358, 2024.

[10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.

[11] R. Caruana, "Multitask learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.

[12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.

[13] P. J. Huber, "Robust estimation of a location parameter," *Annals of Mathematical Statistics*, vol. 35, no. 1, pp. 73–101, 1964.

[14] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proc. 2014 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734.

[15] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[16] O. J. Dunn, "Multiple comparisons among means," *Journal of the American Statistical Association*, vol. 56, no. 293, pp. 52–64, 1961.

[17] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman and Hall/CRC, 1993.

[18] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ, USA: Lawrence Erlbaum Associates, 1988.

[19] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learning Representations (ICLR)*, 2015.

[20] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017, pp. 3319–3328.

[21] P. Zhao, W. Hu, D. Cao, Z. Zhang, W. Liao, Z. Chen, and Q. Huang, "Enhancing multivariate, multi-step residential load forecasting with spatiotemporal graph attention-enabled transformer," *International Journal of Electrical Power & Energy Systems*, vol. 160, p. 110074, 2024.

[22] L. Ø. Bentsen, N. D. Warakagoda, R. Stenbro, and P. Engelstad, "Spatio-temporal wind speed forecasting using graph networks and novel transformer architectures," *Applied Energy*, vol. 333, p. 120565, 2023.